

Pre-training Language Models

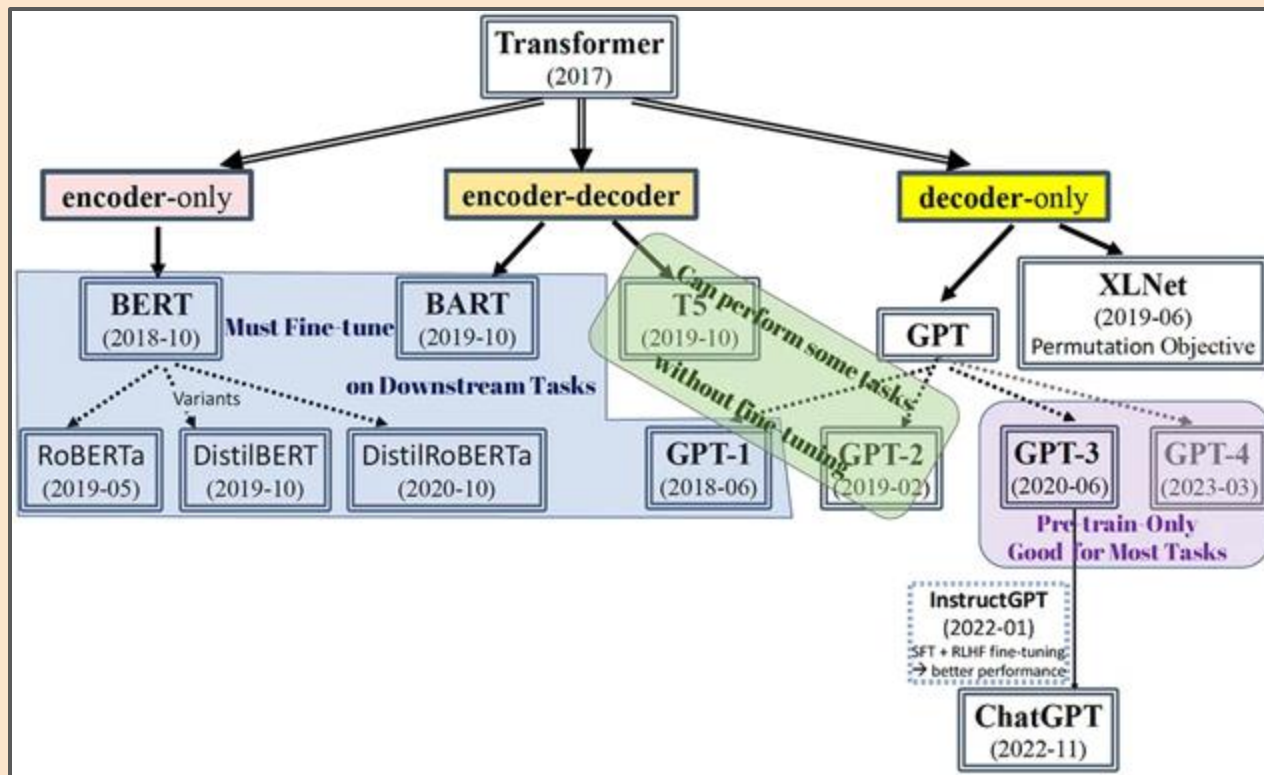


Sourish Dasgupta

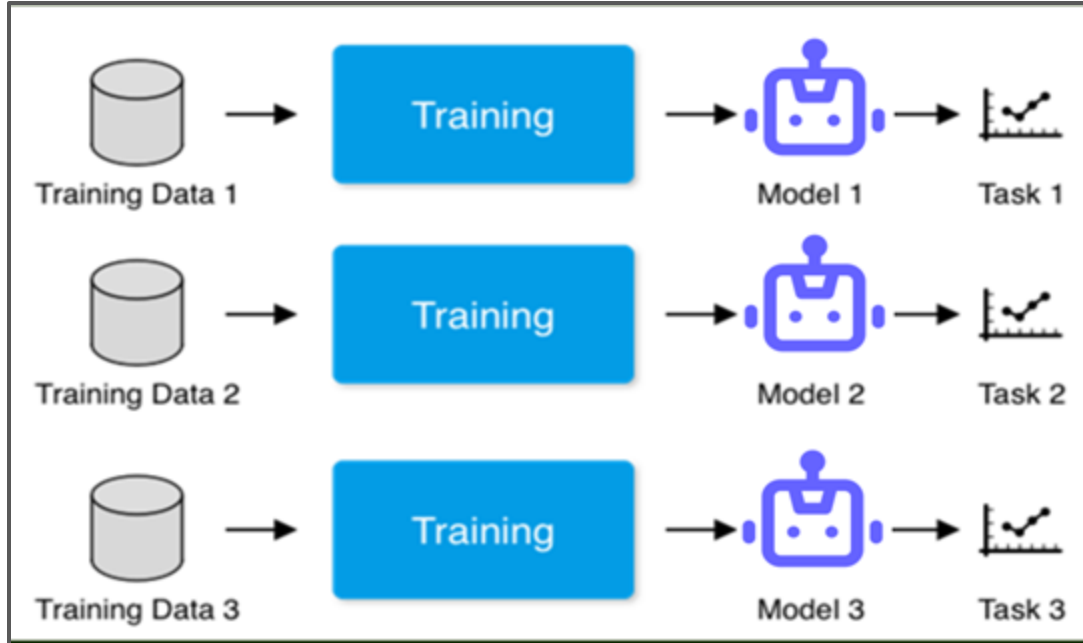
Associate Professor, DAU, Gandhinagar

<https://daiict.ac.in/faculty/sourish-dasgupta>

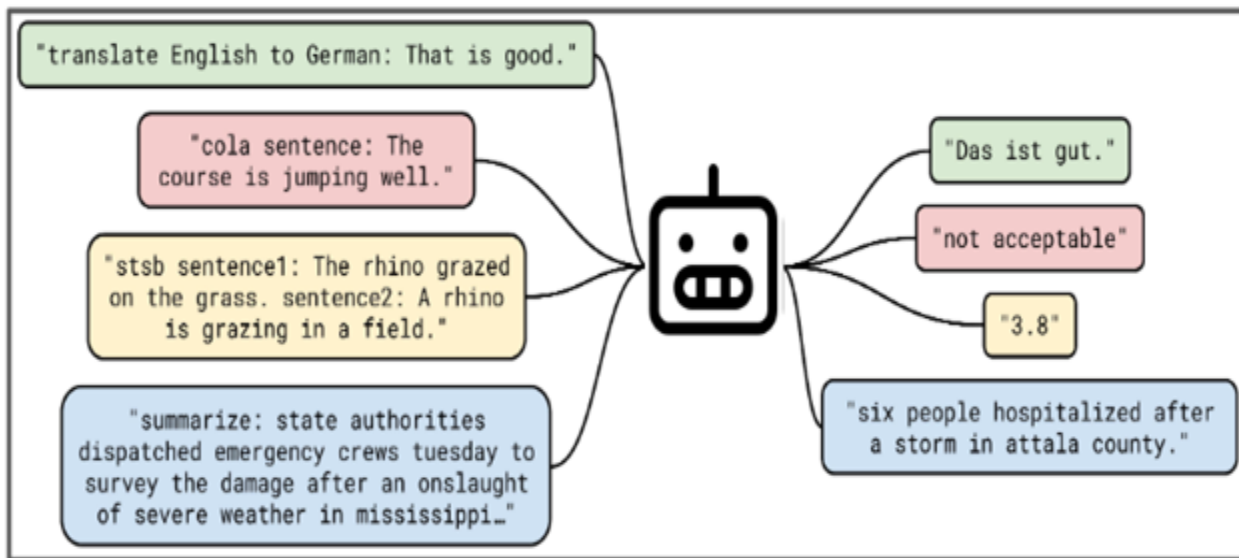
Transformer is the **mother sauce** of most modern LLMs



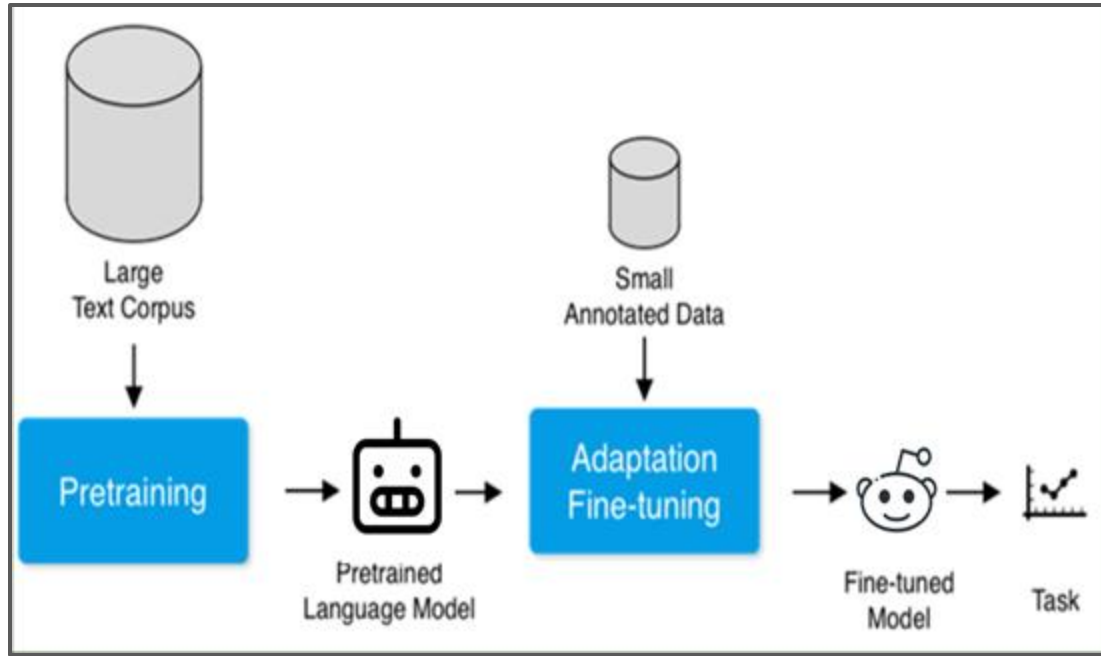
Task-specific Model Training



Pre-training (Multiple Tasks)



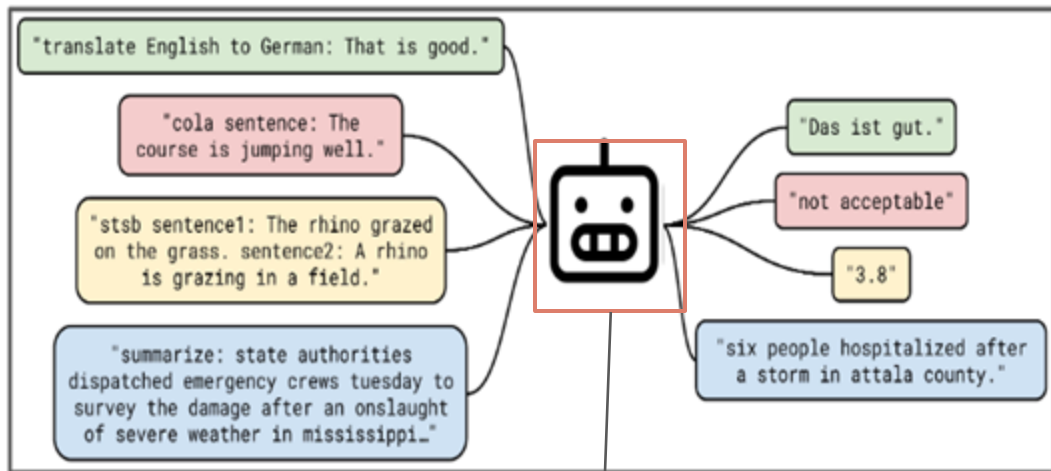
Fine-tuning Pre-trained Models



Three Types of Pre-training Paradigms:

Encoder-Decoder

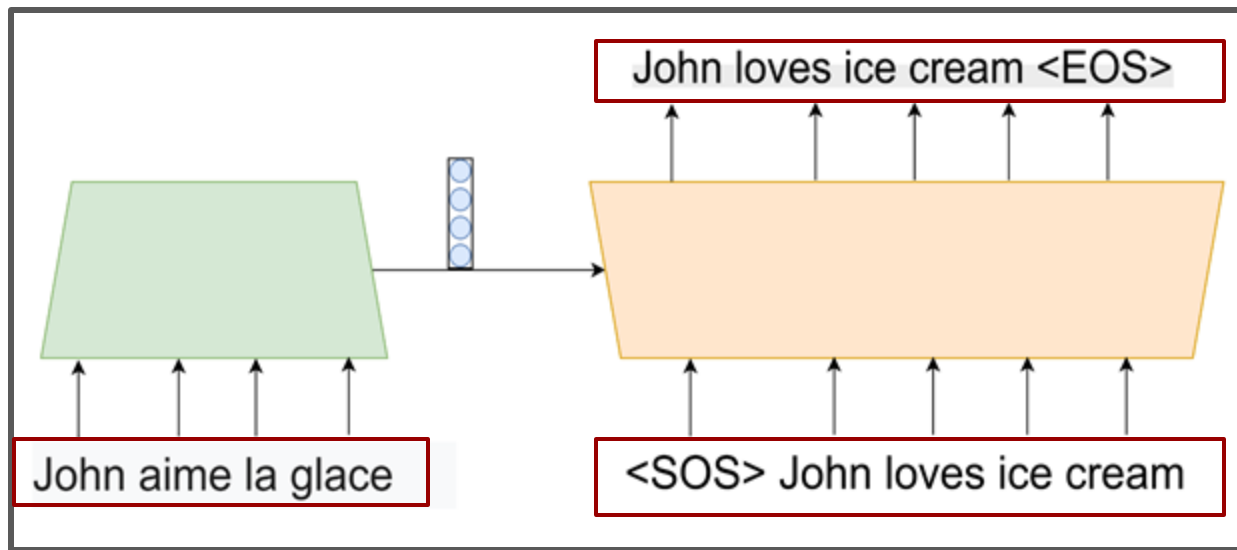
Three Types of Pre-training Paradigms: Encoder-Decoder



Any downstream task can be considered as a Sequence-to-Sequence modeling task

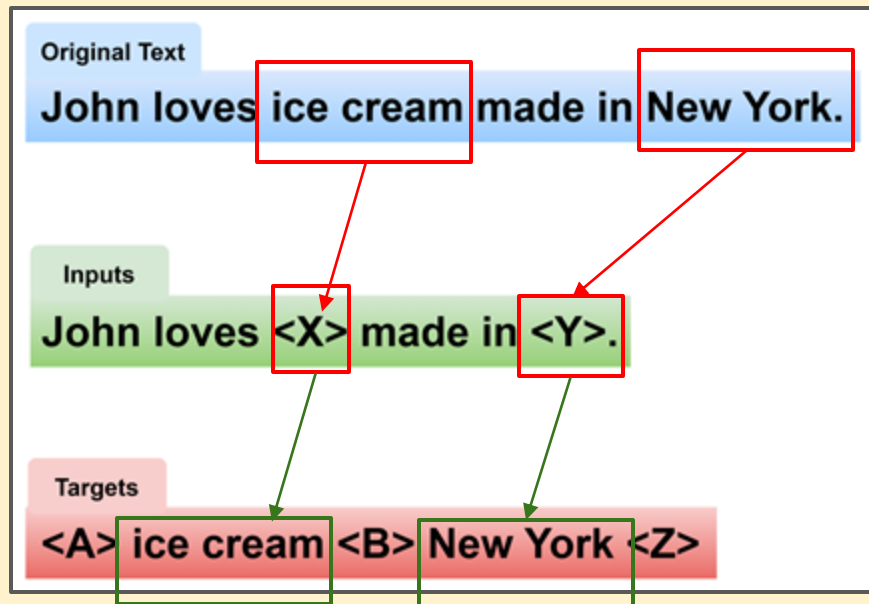
Encoder-Decoder model

Encoder-Decoder Pretraining: Sequence-to-Sequence Task

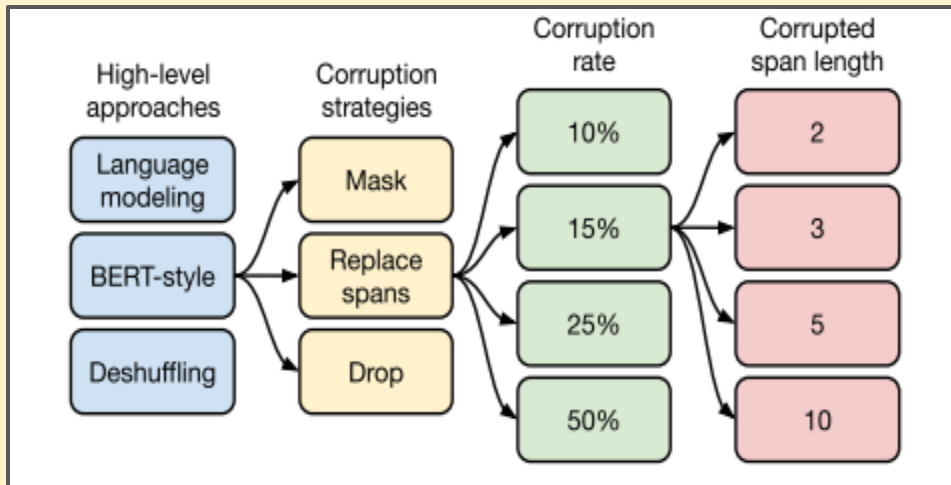


Encoder-Decoder Model: T5

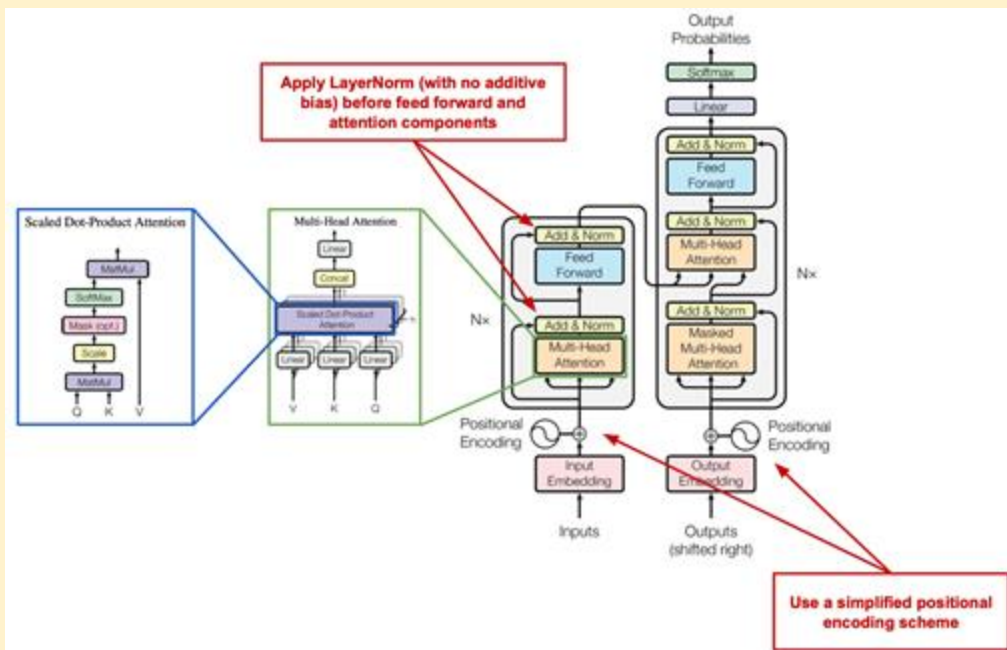
T5 Pre-training Task (Unsupervised): Span Corruption



Training Strategies



T5 Architecture



Pre-training Dataset: Colossal Clean Crawled Corpus

Size & Language:

750GB of cleaned English text from Common Crawl, one of the largest such corpora.

Origin:

Built for the T5 model to overcome limitations of prior datasets (e.g., access, scope, domain bias).

Filtering:

Aggressively cleaned via deduplication, boilerplate removal, and English-only filtering.

Coverage:

Broad web-based content beyond restricted sources like Creative Commons.

Usage:

Default pretraining data for T5; widely used in language modeling and transfer learning.

Performance across Datasets

Model	GLUE Average	CoLA Matthew's	SST-2 Accuracy	MRPC F1	MRPC Accuracy	STS-B Pearson	STS-B Spearman
Previous best	89.4 ^a	69.2 ^b	97.1 ^a	93.6^b	91.5^b	92.7 ^b	92.3 ^b
T5-Small	77.4	41.0	91.8	89.7	86.6	85.6	85.0
T5-Base	82.7	51.1	95.2	90.7	87.5	89.4	88.6
T5-Large	86.4	61.2	96.3	92.4	89.9	89.9	89.2
T5-3B	88.5	67.1	97.4	92.5	90.0	90.6	89.8
T5-11B	90.3	71.6	97.5	92.8	90.4	93.1	92.8

Model	QQP F1	QQP Accuracy	MNLI-m Accuracy	MNLI-mm Accuracy	QNLI Accuracy	RTE Accuracy	WNLI Accuracy
Previous best	74.8 ^c	90.7^b	91.3 ^a	91.0 ^a	99.2^a	89.2 ^a	91.8 ^a
T5-Small	70.0	88.0	82.4	82.3	90.3	69.9	69.2
T5-Base	72.6	89.4	87.1	86.2	93.7	80.1	78.8
T5-Large	73.9	89.9	89.9	89.6	94.8	87.2	85.6
T5-3B	74.4	89.7	91.4	91.2	96.3	91.1	89.7
T5-11B	75.1	90.6	92.2	91.9	96.9	92.8	94.5

Model	SQuAD EM	SQuAD F1	SuperGLUE Average	BoolQ Accuracy	CB F1	CB Accuracy	COPA Accuracy
Previous best	90.1 ^a	95.5 ^a	84.6 ^d	87.1 ^d	90.5 ^d	95.2 ^d	90.6 ^d
T5-Small	79.10	87.24	63.3	76.4	56.9	81.6	46.0
T5-Base	85.44	92.08	76.2	81.4	86.2	94.0	71.2
T5-Large	86.66	93.79	82.3	85.4	91.6	94.8	83.4
T5-3B	88.53	94.95	86.4	89.9	90.3	94.4	92.0
T5-11B	91.26	96.22	88.9	91.2	93.9	96.8	94.8

Model	MultiRC F1a	MultiRC EM	ReCoRD F1	ReCoRD Accuracy	RTE Accuracy	WiC Accuracy	WSC Accuracy
Previous best	84.4 ^d	52.5 ^d	90.6 ^d	90.0 ^d	88.2 ^d	69.9 ^d	89.0 ^d
T5-Small	69.3	26.3	56.3	55.4	73.3	66.9	70.5
T5-Base	79.7	43.1	75.0	74.2	81.5	68.3	80.8
T5-Large	83.3	50.7	86.8	85.9	87.8	69.3	86.3
T5-3B	86.8	58.3	91.2	90.4	90.7	72.1	90.4
T5-11B	88.1	63.3	94.1	93.4	92.5	76.9	93.8

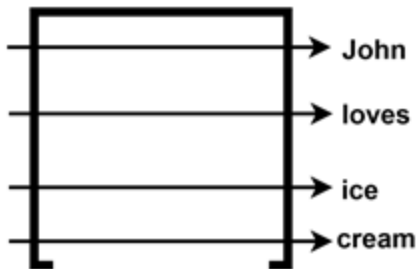
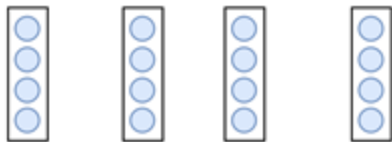
Model	WMT EnDe BLEU	WMT EnFr BLEU	WMT EnRo BLEU	CNN/DM ROUGE-1	CNN/DM ROUGE-2	CNN/DM ROUGE-L
Previous best	33.8^e	43.8^e	38.5^f	43.47 ^g	20.30 ^g	40.63 ^g
T5-Small	26.7	36.0	26.8	41.12	19.56	38.35
T5-Base	30.9	41.2	28.0	42.05	20.34	39.40
T5-Large	32.0	41.5	28.1	42.50	20.68	39.75
T5-3B	31.8	42.6	28.2	42.72	21.02	39.94
T5-11B	32.1	43.4	28.1	43.52	21.55	40.69

Three Types of Pre-training Paradigms:

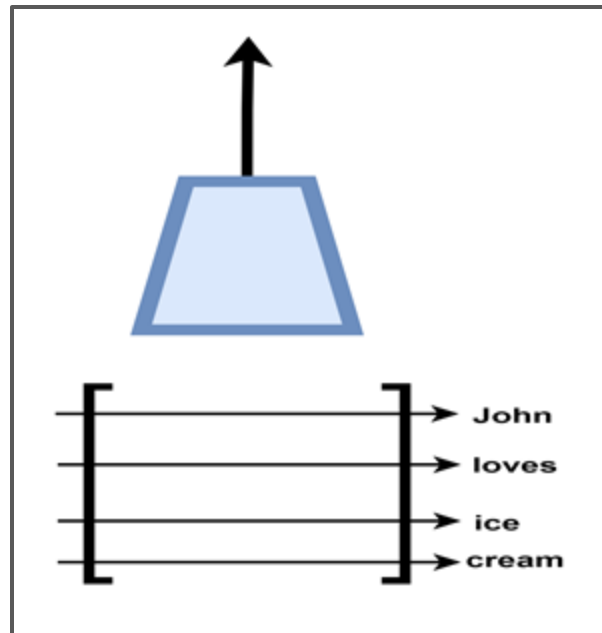
Encoder-Only

Three Types of Pre-training Paradigms: Encoder-Only

John loves ice cream



John loves ice cream



Pre-training Task 1: Masked Token Prediction

Masking → **x% of total**

Type1: Masking with MASK token

John loves ice cream [MASK] in New York.

Type2: Masking with random tokens

John loves ice cream killed in New York.

Type3: Masking with same token

John loves ice cream made in New York.

Pre-training Task 2: Next Sentence Prediction

[CLS] John loves ice cream **[SEP]** He lives in New York **[SEP]**



IsNext

[CLS] John loves ice cream **[SEP]** Eiffel Tower is in Paris **[SEP]**

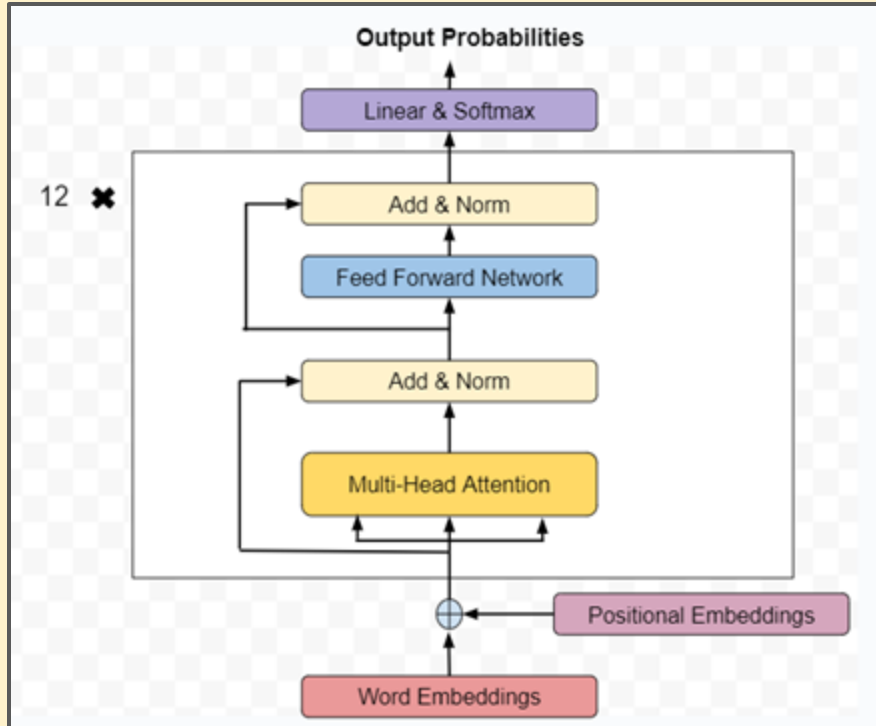


IsNotNext



Encoder-Only Model: BERT

BERT Architecture





Pretraining Dataset: BooksCorpus (800M words)

BooksCorpus (~800 M words)

1. Size & Content

~800 million words from ~11,000 self-published English novels across various genres.

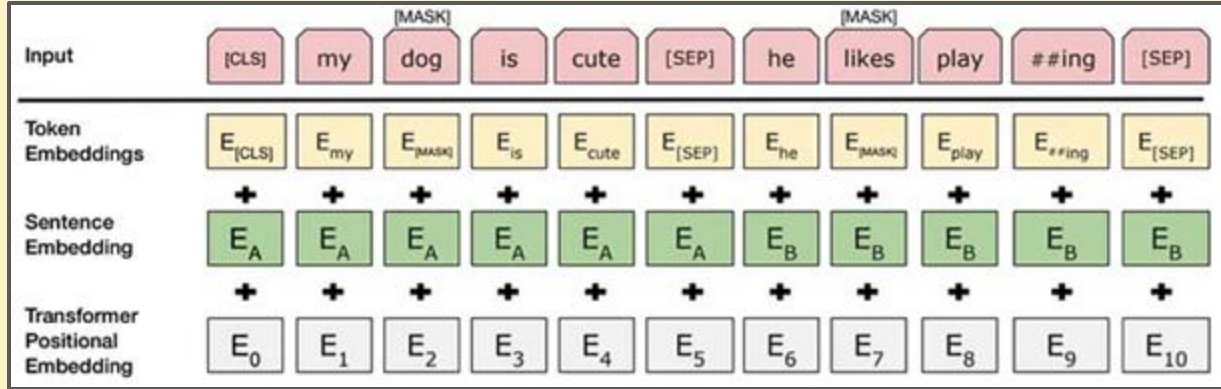
2. Original Purpose

Compiled for aligning books with movies; later adopted for language modeling.

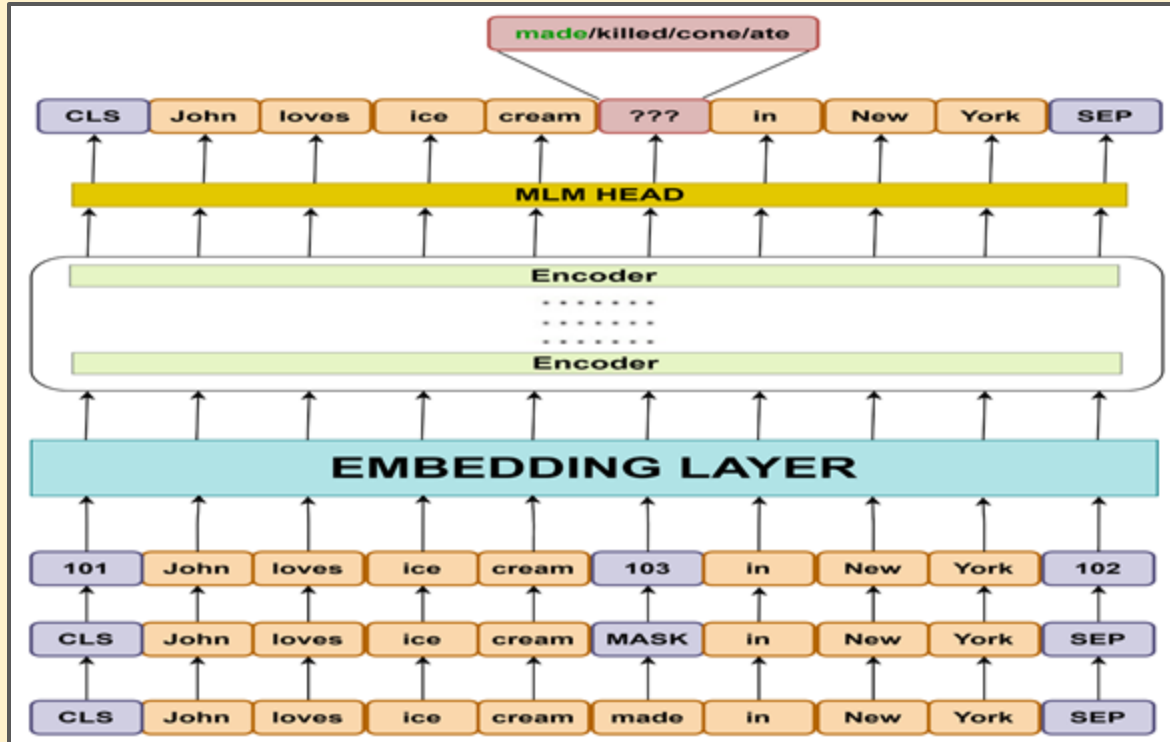
3. Long-form Coherence

Provides extended narrative structures—helpful for training models on story flow and dialogue.

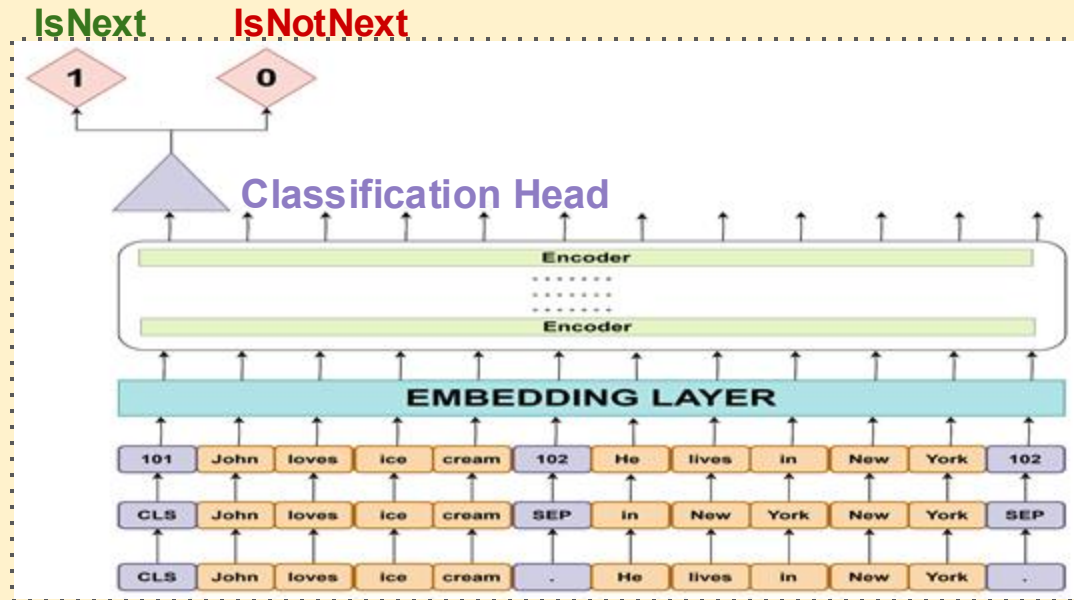
BERT: Initialization of Input



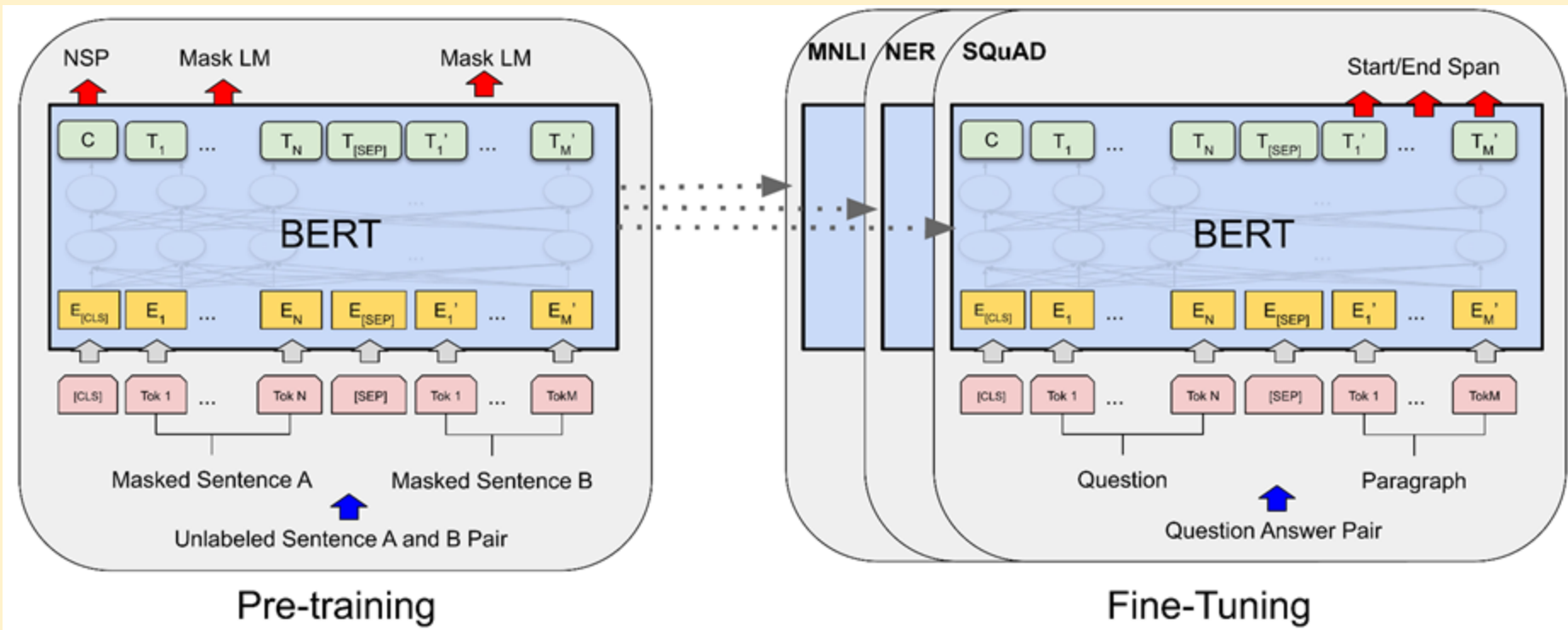
Masked Language Modeling (Pretraining BERT)



BERT NSP Pretraining



BERT: Fine-tuning (more in next lecture)



Performance across Tasks and Datasets



System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

The General Language Understanding Evaluation (GLUE) benchmark (Wang et al., 2018a) is a collection of diverse natural language understanding tasks.



Performance: Q/A tasks

SQuAD 1

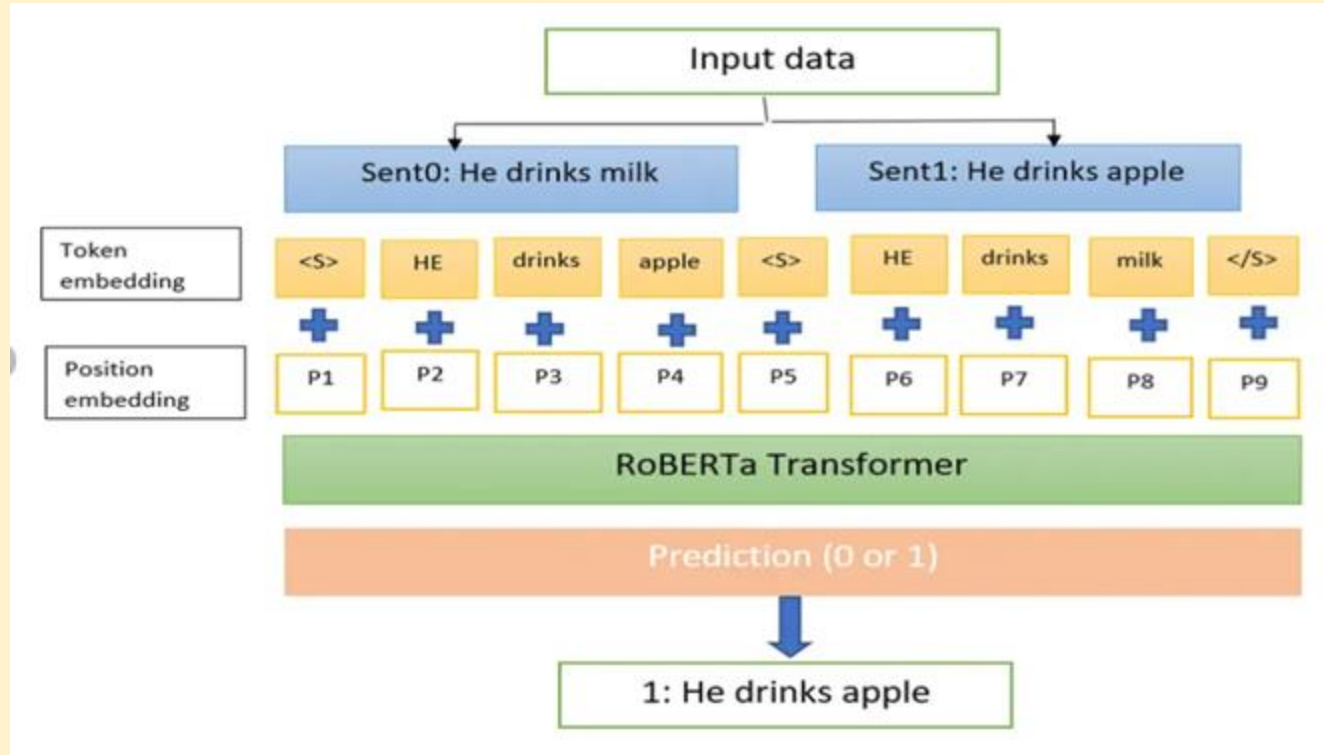
SQuAD 2

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	-	-	82.3	91.2
#1 Ensemble - nlnet	-	-	86.0	91.7
#2 Ensemble - QANet	-	-	84.5	90.5
Published				
BiDAF+ELMo (Single)	-	85.6	-	85.8
R.M. Reader (Ensemble)	81.2	87.9	82.3	88.5
Ours				
BERT _{BASE} (Single)	80.8	88.5	-	-
BERT _{LARGE} (Single)	84.1	90.9	-	-
BERT _{LARGE} (Ensemble)	85.8	91.8	-	-
BERT _{LARGE} (Sgl.+TriviaQA)	84.2	91.1	85.1	91.8
BERT _{LARGE} (E2S.+TriviaQA)	86.2	92.2	87.4	93.2

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	86.3	89.0	86.9	89.5
#1 Single - MIR-MRC (F-Net)	-	-	74.8	78.0
#2 Single - nlnet	-	-	74.2	77.1
Published				
unet (Ensemble)	-	-	71.4	74.9
SLQA+ (Single)	-	-	71.4	74.4
Ours				
BERT _{LARGE} (Single)	78.7	81.9	80.0	83.1

The Stanford Question Answering Dataset (SQuAD v1.1) is a collection of 100k crowd-sourced question/answer pairs (Rajpurkar et al.,

Variations of BERT: RoBERTa



Three Types of Pre-training Paradigms:

Decoder-Only

Decoder-Only Models are Causal Language Models (CLM)

John loves ice cream made in New York.

John

loves

John loves

ice

John loves ice

cream

John loves ice cream

made

John loves ice cream made

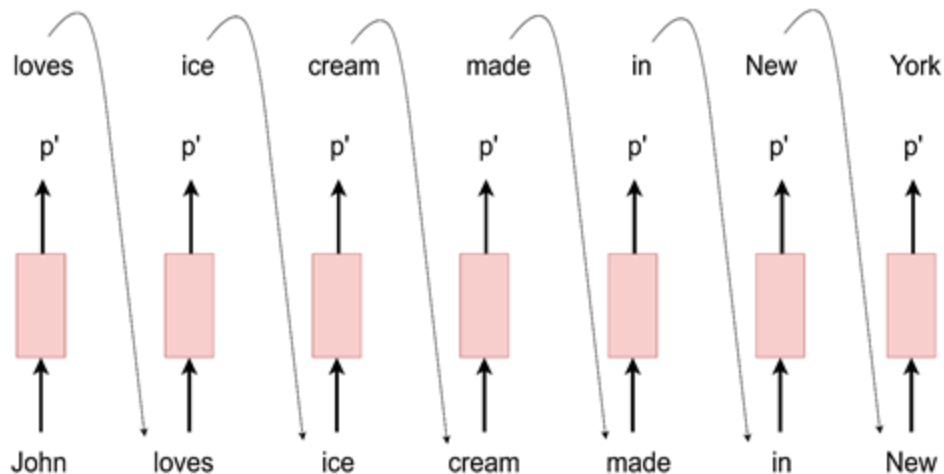
Forward-masking based Training

in

John loves ice cream made in

New

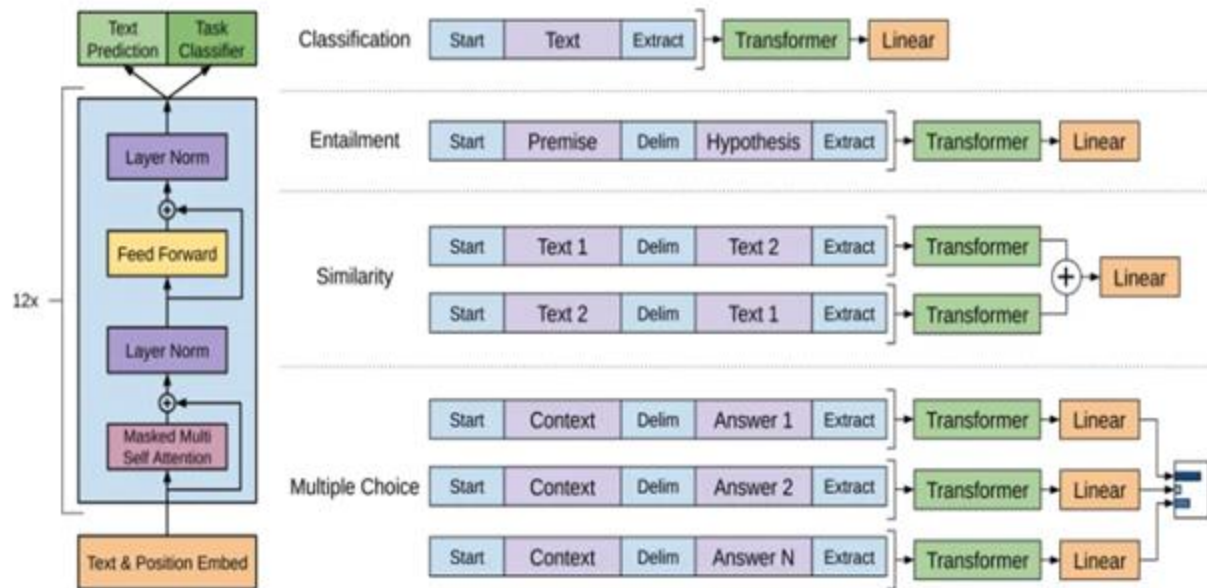
Decoder-Only Models are Causal Language Models (CLM)



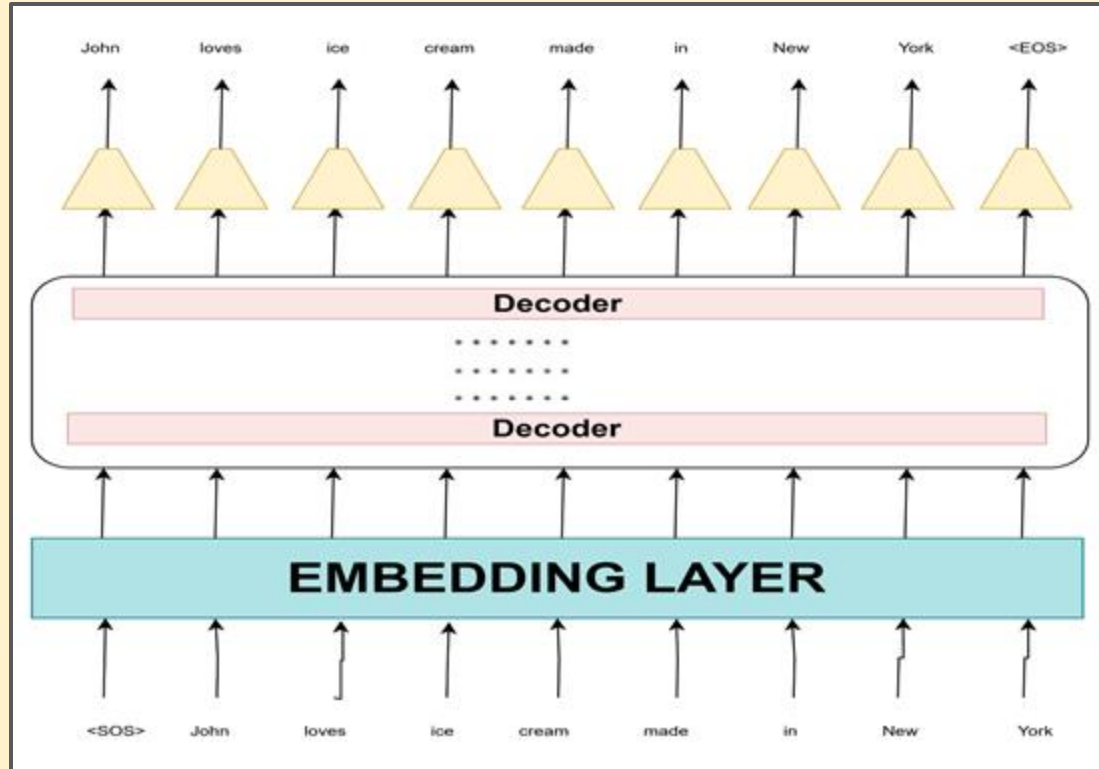
Teacher-forcing Training

Decoder-Only Model: GPT

GTP Architecture



Pretraining GPT : CLM



BooksCorpus (~800 M words)

1. Size & Content

~800 million words from ~11,000 self-published English novels across various genres.

2. Original Purpose

Compiled for aligning books with movies; later adopted for language modeling.

3. Long-form Coherence

Provides extended narrative structures—helpful for training models on story flow and dialogue.

Evaluation

Task	Datasets
Natural language inference	SNLI [5], MultiNLI [66], Question NLI [64], RTE [4], SciTail [25]
Question Answering	RACE [30], Story Cloze [40]
Sentence similarity	MSR Paraphrase Corpus [14], Quora Question Pairs [9], STS Benchmark [6]
Classification	Stanford Sentiment Treebank-2 [54], CoLA [65]

Evaluation

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0

Table 3: Results on question answering and commonsense reasoning, comparing our model with current state-of-the-art methods.. 9x means an ensemble of 9 models.

Method	Story Cloze	RACE-m	RACE-h	RACE
val-LS-skip [55]	76.5	-	-	-
Hidden Coherence Model [7]	<u>77.6</u>	-	-	-
Dynamic Fusion Net [67] (9x)	-	55.6	49.4	51.2
BiAttention MRU [59] (9x)	-	<u>60.2</u>	<u>50.3</u>	<u>53.3</u>
Finetuned Transformer LM (ours)	86.5	62.9	57.4	59.0